

XAUUSD Strategy

Internal Research Summary: Introduction, Motivation, Mathematical Model, and Test Data

XAUUSD Modelling Research Team

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Abstract

This is an internal research document for team use. It records what was tried, what failed, what survived, and what should be tested next. It deliberately removes nonessential diagrams and focuses on the introduction, motivation, mathematical structure, test data, implementation status, and empirical interpretation. The strategy is a regime-conditioned gold model, but the validated evidence is narrower than the full conceptual architecture. Real 4H XAUUSD data does not validate a 75% directional hit-rate target with the current price-only feature set. The durable evidence supports a long-only, trend-regime gold strategy with realistic forward expectations around 48–55% hit rate and Sharpe 1.5–2.5, pending forward paper validation and macro-factor expansion.

Internal Research Document. Classification: for team use only. Status: Phase 6 complete, Phase 7 planning. This document is a research summary and knowledge-transfer artifact, not an external paper, marketing document, or production deployment approval.

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1 Document Control

Table 1: Document control and intended use.

Field	Value
Classification	Internal research document; for team use only.
Primary purpose	Preserve research findings, negative results, mathematical assumptions, test evidence, and next actions.
Audience	Quant research, engineering, execution/risk, and leadership stakeholders deciding Phase 7 scope.
Status	Phase 6 complete; Phase 7 planning.
Deployment meaning	Not a live-trading approval. The current strategy is a paper-trading candidate only.
Core decision recorded	Abandon the original 75% hit-rate target for the current price-only 4H feature set; continue with long-only trend-regime paper trading and macro-factor research.

This document should be read as a research ledger. It records both successes and failures so future work does not repeat the same experiments without adding new information.

2 Introduction

XAUUSD is the spot price of gold quoted in U.S. dollars. It is not simply another foreign-exchange pair. Gold trades as a monetary asset, a real-rate asset, a safe-haven asset, an inflation hedge, and a liquidity-sensitive commodity. A strategy that treats it as a generic short-horizon momentum series will usually overstate edge in backtests and underperform when macro regimes change.

The strategy studied here is a hybrid regime model for gold. Its purpose is not to predict every bar. Its purpose is to answer a narrower question:

When does gold enter a market state where long exposure has positive expected value after transaction costs, drawdown controls, and out-of-sample decay?

The answer emerging from the research is that broad 4H direction prediction is weak, but filtered long-only trend participation is viable. In the cleanest pre-registered test, a long-only trend-regime strategy produced a 53.0% hit rate with a bootstrap hit-rate confidence interval of 46.6–59.4%. Its Monte Carlo Sharpe median was 1.97 with a 95% interval of 0.89–2.78. That does not justify an aggressive live deployment, but it does justify continued paper trading and event/macro feature expansion.

3 Motivation

3.1 Why Gold Needs a Regime Model

Gold changes behavior across macro and liquidity states. The same 4H candle pattern can mean different things depending on whether the market is in a quiet Asian session, a London trend discovery window, a New York macro impulse, a risk-off squeeze, or a liquidity vacuum. A stationary model assumes one stable mapping from features to returns. That assumption is too strong for XAUUSD.

The regime model addresses this by making the conditional law of returns state dependent. Informally,

future return law = function of current regime, flow state, macro forcing, and execution conditions.

The target is not a single universal alpha rule. The target is a rule that can say “trade”, “trade smaller”, or “do not trade” depending on which market is active.

3.2 Why the Original High-Hit-Rate Target Failed

The original target was a hit rate above 75% and Sharpe in the 2.5–3.3 range. The real-data methodology report shows that this target is not validated on current 4H price-only data:

- baseline 4H momentum reached only 51.6%;
- mean reversion after negative bars reached 53.3%;
- post-FOMC drift improved to 56.9%, showing that events matter;
- ML top 3% selections reached 75.3% in sample but collapsed to 50.5% walk-forward.

The failure is therefore not mainly an optimizer failure. It is an information failure. Gold is driven by FOMC repricing, CPI surprises, NFP shocks, DXY, real yields, risk-off flows, and session liquidity. If those variables are absent, the model is trying to infer macro information from price alone.

3.3 What Actually Survived

The valid insight is:

Long-only gold exposure performs better when restricted to stronger trend regimes, with conservative parameters chosen before testing.

This is different from saying that the model can forecast the next bar with 75% accuracy. The surviving strategy is closer to controlled participation: avoid shorting gold, require trend structure, use volatility-normalized exits, and control risk through drawdown and loss-streak rules.

4 Key Research Decisions

Table 2: Decision record for the current research phase.

Decision	Status	Rationale
Retire the 75% target for the current feature set	Accepted	Real walk-forward tests do not support 75% hit rate from 4H price-only features. The only 75% result was in-sample and collapsed out of sample.
Keep long-only trend-regime candidate	Accepted	The strongest repeatable evidence is controlled participation in long gold trend regimes, not symmetric long/short prediction.
Treat risk overlay as survival technology	Accepted	Drawdown and ruin improved, but negative-EV cohorts were not transformed into positive-EV cohorts. Risk controls do not create alpha.
Treat continuum model as architecture, not proof	Accepted	The continuum equations organize intuition around liquidity, flow, friction, and forcing. They are not themselves empirical validation.
Make macro/event data the next research branch	Accepted	Post-FOMC drift outperformed generic price-only rules, and the largest missing variables are macro drivers directly relevant to gold.
Paper trade before any live deployment	Required	Sample size, regime concentration, data provenance, and cost assumptions require forward validation.

5 Strategy Overview

The strategy has four layers:

1. **State estimation:** infer the current market state and regime.
2. **Signal construction:** combine regime posterior, trend, flow, and macro forcing into a score.
3. **Trade admission:** require threshold, confidence, drift, spread, and venue-health conditions.
4. **Risk overlay:** scale down or stop trading when drawdown or loss streaks indicate deterioration.

The validated base candidate is intentionally simple:

Table 3: Validated base candidate.

Component	Setting
Direction	Long only
Trend filter	$ADX \geq 25$
EMA slope filter	≥ 0.0001
Stop distance	$2 \times ATR$
Target distance	$4 \times ATR$
Nominal reward/risk	2 : 1
Shorting	Disabled
Position sizing in reported test	1% risk per trade

The risk overlay studied later uses a lower risk fraction, approximately 0.493% per trade, plus drawdown and loss-streak controls. That overlay is defensive: it improves survival metrics but does not create predictive edge.

6 Implementation Status

The full mathematical architecture is broader than the code path that generated the cleanest empirical evidence. This distinction matters. The table below separates implemented research artifacts from conceptual or next-phase components.

Table 4: Implemented versus conceptual components.

Component	Status	Notes
Long-only trend filter	Implemented and tested	This is the core pre-registered regime-filtered candidate: ADX, EMA slope, breakout, 2ATR stop, 4ATR target.
Risk overlay	Implemented and tested separately	Drawdown ladder, half-size rule, skip rule, catastrophe cap, and reduced risk fraction were tested through Monte Carlo overlays.
Walk-forward validation	Implemented	Used to reject in-sample 75% ML results and enforce out-of-sample interpretation.
Purged embargo	Implemented	A 12-bar embargo was used to reduce overlap and indicator leakage.
HMM regime filter	Conceptual or partial research layer	Included as the intended mathematical state-estimation structure; not the source of the cleanest validated backtest result.
Continuum liquidity-flow model	Conceptual architecture	Useful for intuition and future feature design. It should not be presented as an empirically calibrated PDE model.
Macro factor integration	Not yet tested in the final candidate	FOMC, CPI, NFP, DXY, real yields, and risk-off variables are the recommended Phase 7 branch.
Live execution stack	Not approved	Current evidence supports paper trading and further research, not live capital deployment.

7 Mathematical Setting

7.1 Filtered Probability Space

Let

$$(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t \geq 0}, \mathbb{P})$$

be a filtered probability space. The sigma-algebra $\mathcal{F}_t$ contains all information available up to time t : XAUUSD quotes, gold futures, OHLCV bars, derived features, session labels, macro-event flags, historical fills, and model diagnostics.

The trading decision at time t must be $\mathcal{F}_t$ -measurable. This is the mathematical way of saying there must be no lookahead. Any feature used for entry at time t must be observable at or before t .

7.2 Observed Data

Let S_t be the observed XAUUSD or gold proxy price at time t . Define the log return

$$r_t = \log S_t - \log S_{t-1}.$$

For a bar series, typical derived variables include:

$$ATR_t, \quad ADX_t, \quad EMA_t, \quad \Delta EMA_t, \quad Vol_t, \quad Session_t.$$

The base candidate uses trend and volatility features. The future extension should add macro forcing:

$$M_t = (r_t^{GC}, \Delta DXY_t, \Delta y_{2y,t}^{real}, \Delta y_{10y,t}^{real}, CPI_t, NFP_t, FOMC_t, RiskOff_t, Session_t)^\top.$$

7.3 Latent Regime

Let the hidden market regime be

$$Z_t \in \mathcal{Z} = \{1, \dots, K\}.$$

The regime posterior is

$$\pi_t^i = \mathbb{P}(Z_t = i \mid \mathcal{F}_t), \quad \sum_{i=1}^K \pi_t^i = 1.$$

The posterior is more informative than a hard label because markets often sit between regimes.

In discrete time, if A_t is the transition matrix and $\ell_t^i = p(Y_t \mid Z_t = i, \mathcal{F}_{t-1})$ is the emission likelihood, the hidden Markov filter is

$$\begin{aligned} \tilde{\pi}_t^i &= \sum_{j=1}^K \pi_{t-1}^j A_t(j, i), \\ \pi_t^i &= \frac{\ell_t^i \tilde{\pi}_t^i}{\sum_{k=1}^K \ell_t^k \tilde{\pi}_t^k}. \end{aligned}$$

This is a prediction-update recursion. The previous posterior is pushed through the transition matrix, then reweighted by how likely the current observation is under each regime.

8 Reduced State and Flow Interpretation

8.1 State Vector

The full conceptual model uses the reduced state

$$X_t = (m_t, u_t, I_t, \rho_t, s_t, \sigma_t, b_t)^\top,$$

where:

Table 5: State variables.

Variable	Name	Explanation
m_t	latent fair price	A filtered estimate of local fair value. Observed spot price is $S_t = m_t + \varepsilon_t$.
u_t	price-flow velocity	Direction and strength of local price movement.
I_t	signed imbalance	Net buy/sell pressure from order flow and bar features.
ρ_t	liquidity density	Effective ability of the market to absorb orders.
s_t	spread/friction	Execution cost and market-health proxy.
σ_t	local volatility	Used for stops, sizing, and uncertainty.
b_t	spot-futures basis	Difference or lead-lag relation between spot XAUUSD and GC futures.

The base backtest does not fully estimate all of these variables in production-grade form. The state vector describes the intended architecture and gives a disciplined way to extend the strategy beyond price-only features.

8.2 Continuum Prior

The continuum prior is a modeling analogy. It says that price moves through a market medium with liquidity, imbalance, friction, and external forcing. Define fields over an abstract coordinate x :

$$p(x, t), \quad u(x, t), \quad \rho(x, t), \quad I(x, t), \quad \Pi(x, t), \quad \sigma(x, t).$$

The deterministic skeleton is:

$$\partial_t \rho + \nabla \cdot (\rho u) = s_\rho, \tag{1}$$

$$\rho (\partial_t u + (u \cdot \nabla) u) = -\nabla \Pi + \nabla \cdot \tau + \rho f, \tag{2}$$

$$\partial_t p + u \cdot \nabla p = \kappa \Delta p + g, \tag{3}$$

$$\partial_t I + \nabla \cdot (I u) = D_I \Delta I - \alpha I + \eta, \tag{4}$$

$$\Pi = a \rho^\gamma + b I + c \sigma, \tag{5}$$

$$\tau = \mu (\nabla u + (\nabla u)^\top) + \lambda (\nabla \cdot u) \mathbf{I}. \tag{6}$$

Equation (1) is liquidity conservation with entry/withdrawal s_ρ . Equation (2) says pressure gradients and forcing move price-flow velocity. Equation (3) says price is transported by velocity and smoothed by diffusion. Equation (4) says imbalance is itself transported and mean-reverting. Equation (5) defines market pressure as a function of liquidity, imbalance, and volatility.

The analogy should not be overclaimed. It is useful because it forces every trade signal to be explainable in terms of market state: liquidity, imbalance, volatility, macro forcing, and friction.

8.3 Reduced Stochastic Dynamics

In regime $i = Z_t$, the reduced dynamics can be written:

$$dm_t = (\alpha_i u_t + \beta_i I_t + \gamma_i r_t^{GC} + \delta_i^\top M_t) dt + \sigma_{m,i} dW_t^{(m)}, \tag{7}$$

$$du_t = (-\lambda_i u_t - \partial_x \Pi_{i,t} + \nu_i \partial_{xx} u_t + \varphi_i^\top M_t) dt + \sigma_{u,i} dW_t^{(u)}, \tag{8}$$

$$dI_t = (-\phi_i I_t + \psi_i O F I_t + \chi_i^\top M_t) dt + \sigma_{I,i} dW_t^{(I)}, \tag{9}$$

$$d\rho_t = (-\partial_x (\rho_t u_t) - c_i \rho_t + d_i Q A_t) dt + \sigma_{\rho,i} dW_t^{(\rho)}, \tag{10}$$

$$ds_t = a_i (\bar{s}_i - s_t) dt + b_i^s Vol Shock_t dt + \sigma_{s,i} dW_t^{(s)}, \tag{11}$$

$$d\sigma_t = \kappa_i (\bar{\sigma}_i - \sigma_t) dt + \eta_i |r_t| dt + \sigma_{\sigma,i} dW_t^{(\sigma)}, \tag{12}$$

$$db_t = -\zeta_i b_t dt + \omega_i (r_t^{spot} - r_t^{GC}) dt + \sigma_{b,i} dW_t^{(b)}. \tag{13}$$

Each equation has a practical interpretation:

- fair value changes with velocity, imbalance, futures lead, and macro forcing;
- velocity decays unless pressure or macro forcing sustains it;
- imbalance mean-reverts but is replenished by order-flow imbalance;
- liquidity density can be depleted by aggressive flow or replenished by quoted activity;
- spread and volatility mean-revert but jump during shocks;
- the spot-futures basis mean-reverts but widens when spot and futures disagree.

9 Signal Construction

9.1 Regime-Specific Score

A regime-specific base score is:

$$g_i^{base}(X_t, M_t) = w_{1,i}\hat{u}_t + w_{2,i}\hat{I}_t + w_{3,i}r_t^{GC} - w_{4,i}\Delta DXY_t - w_{5,i}\Delta y_t^{real} \\ + w_{6,i}RiskOff_t - w_{7,i}s_t - w_{8,i}|b_t|.$$

The signs encode gold intuition. Positive price-flow velocity, positive imbalance, and positive gold-futures lead support long exposure. Rising DXY and rising real yields usually pressure gold, so they enter with negative signs. High spreads and large basis dislocations reduce signal quality.

The posterior-weighted aggregate score is

$$Score_t = \sum_{i=1}^K \pi_t^i g_i^{eff}(X_t, M_t).$$

If an adaptive observer is used, it can add a clipped adjustment:

$$g_i^{eff}(X_t, M_t, h_t) = g_i(X_t, M_t; w_{i,t}^{eff}) + \text{clip}(\Delta g_{i,t}(h_t), B_{g,i}).$$

The clipping is important. It prevents the adaptive layer from dominating the interpretable base model.

9.2 Base Rule Used in the Clean Test

The cleanest empirical test does not rely on the full adaptive score. It uses a conservative long-only trend rule:

$$A_t^{long} = \mathbf{1}\{ADX_t \geq 25, \Delta EMA_t \geq 0.0001, Breakout_t = 1\}.$$

If $A_t^{long} = 1$, the system enters a long position with:

$$StopDistance_t = 2 \cdot ATR_t, \quad TargetDistance_t = 4 \cdot ATR_t.$$

This corresponds to a nominal reward/risk ratio of

$$\frac{TargetDistance_t}{StopDistance_t} = 2.$$

10 Risk and Position Sizing

10.1 Volatility-Normalized Sizing

A generic volatility-normalized size is

$$q_t^{base} = \frac{RiskBudget \cdot C_t \cdot |Score_t|}{\hat{\sigma}_t \cdot StopDistance_t}.$$

Here C_t is confidence and $\hat{\sigma}_t$ is local volatility. The denominator penalizes wider stops and higher volatility.

10.2 Drawdown Ladder

The selected risk-overlay candidate uses a drawdown ladder:

$$L(d) = \begin{cases} 1.00, & d < 0.06, \\ 0.85, & 0.06 \leq d < 0.10, \\ 0.75, & 0.10 \leq d < 0.14, \\ 0.50, & 0.14 \leq d < 0.18, \\ 0.00, & d \geq 0.20. \end{cases}$$

The practical meaning is simple: as drawdown increases, size decreases; at the stop tier, new trading is disabled.

10.3 Loss-Streak Controls

Let

$$\ell_t = \mathbf{1}\{\text{PnL}_t < 0\}.$$

The half-size rule is activated when at least three of the last five trades are losses:

$$H_t = \begin{cases} 0.5, & \sum_{k=t-4}^t \ell_k \geq 3, \\ 1.0, & \text{otherwise.} \end{cases}$$

The skip rule is activated when at least four of the last six trades are losses:

$$S_t^{skip} = \begin{cases} 0, & \sum_{k=t-5}^t \ell_k \geq 4, \\ 1, & \text{otherwise.} \end{cases}$$

The executed size is then

$$q_t = q_t^{base} \cdot L(\text{DD}_t) \cdot H_t \cdot S_t^{skip}.$$

11 Performance Mathematics

11.1 Hit Rate and Expectancy

Let p be hit rate, W average win, and $L > 0$ average absolute loss. Per-trade expectancy is:

$$\mu_{trade} = pW - (1 - p)L.$$

The break-even hit rate is:

$$p^* = \frac{L}{W + L} = \frac{1}{1 + W/L}.$$

For example, the promising gate-exit experiment reported win/loss ratio $W/L = 2.23$. Its break-even hit rate is:

$$p^* = \frac{1}{1 + 2.23} \approx 30.96\%.$$

This explains why improved exit geometry matters: a strategy can survive a modest hit rate if its payoff ratio is strong. However, payoff ratio alone does not solve sample-size and regime-concentration problems.

11.2 Sharpe Ratio

For trade returns $R_1, \dots, R_n$, define:

$$\hat{\mu} = \frac{1}{n} \sum_{t=1}^n R_t, \quad \hat{\sigma}^2 = \frac{1}{n-1} \sum_{t=1}^n (R_t - \hat{\mu})^2.$$

If N is the expected annual trade count, a trade-level annualized Sharpe estimate is:

$$\widehat{\text{Sharpe}} = \sqrt{N} \frac{\hat{\mu}}{\hat{\sigma}}.$$

This estimate becomes unstable when trade count is low or returns are clustered. That is why the final interpretation relies on confidence intervals and forward paper trading, not only point estimates.

11.3 Drawdown and Ruin

For an equity curve E_t , drawdown is:

$$\text{DD}_t = 1 - \frac{E_t}{\max_{0 \leq s \leq t} E_s}.$$

Maximum drawdown over a path is:

$$\text{MDD} = \max_{0 \leq t \leq T} \text{DD}_t.$$

If E_{ruin} is a ruin threshold, probability of ruin over B Monte Carlo paths is:

$$\widehat{\mathbb{P}}(\text{ruin}) = \frac{1}{B} \sum_{b=1}^B \mathbf{1} \left\{ \min_t E_t^{(b)} \leq E_{\text{ruin}} \right\}.$$

12 Test Data

12.1 Primary Real-Data Substrate

The primary real-data analysis used Kaggle XAUUSD 1-minute data:

Table 6: Primary XAUUSD test data.

Item	Value
Source	Kaggle XAUUSD 1-minute data
Raw coverage	2004–2026
Raw rows	approximately 6.8 million
Research bar size	4H bars
Primary period analyzed	2018–2020
Extended analysis	through 2025 where available
Core purpose	test whether 4H price features can reach high hit rate

The 1-minute data was resampled to 4H bars. This standardizes the research horizon and reduces microstructure noise, but it also creates a key limitation: many macro-event moves happen inside a 4H bar. Without event labels, the model sees the price response but not the event cause.

12.2 Data Provenance and Quality Note

The Kaggle 1-minute dataset is useful for research because it gives broad historical coverage and enough density to construct 4H bars. It is not, by itself, institutional-grade execution data. Before any live deployment decision, the team should independently verify:

- timestamp timezone and daylight-saving handling;
- missing bars and duplicated bars;
- abnormal spikes, stale quotes, and weekend prints;
- bid/ask availability versus midpoint-only assumptions;

- broker/feed differences versus the intended execution venue;
- whether gold futures, spot XAUUSD, and macro timestamps are aligned without lookahead.

The current writeup treats the Kaggle data as sufficient for rejecting overfit claims and identifying research direction. It does not treat the dataset as sufficient for production execution approval.

12.3 Backtesting Standards

The methodology report applied:

- walk-forward validation with train/validation/test separation;
- a 12-bar purged embargo to reduce lookahead contamination;
- regularized ML models and conservative thresholds;
- out-of-sample validation of hit-rate claims;
- comparison of synthetic-data results against real-data behavior.

The purged embargo matters because adjacent bars overlap in information. If the training set ends immediately before the test set, leakage can enter through indicators, event effects, or horizon overlap. The embargo creates a gap:

$$Train = [t_0, t_1], \quad Embargo = (t_1, t_1 + 12], \quad Test = (t_1 + 12, t_2].$$

12.4 Strategies Tested on Real 4H Data

Table 7: Real 4H strategy tests from the methodology report.

Strategy	Trades	Hit rate	Interpretation
Baseline momentum	2368	51.6%	Near random directional edge.
Mean reversion after negative bar	1444	53.3%	Best simple 4H rule.
London momentum	371	51.5%	No meaningful edge.
New York momentum	393	45.5%	Negative edge.
Post-FOMC drift	144	56.9%	Events help, but labels are needed.
ML top 3% in sample	93	75.3%	Overfit.
ML top 3% walk-forward	93	50.5%	Fails to generalize.

The key empirical fact is that the simple 4H feature set tops out around 53% hit rate. A 75% target requires new information, not just a better threshold search.

12.5 Pre-Registered Regime Test Data

The pre-registered regime-filtered momentum report used fixed parameters before testing:

Table 8: Pre-registered regime test summary.

Metric	Value
Total trades	219
Hit rate	53.0%
Hit-rate 95% CI	46.6–59.4%
Point-estimate Sharpe	3.87
Monte Carlo Sharpe median	1.97
Sharpe 95% CI	0.89–2.78
Max drawdown	8.2%
Realistic live hit-rate target	48–55%
Realistic live Sharpe target	1.5–2.5

The point-estimate Sharpe is not treated as the forward expectation. The Monte Carlo interval and the 2024 outlier warning reduce the practical claim. The report explicitly notes that 2024 produced 80% hit rate on only 20 trades and should be treated as noise.

12.6 Yearly Performance

Table 9: Yearly performance in the pre-registered test.

Year	Trades	Hit rate	PnL	Comment
2018	22	40.9%	49	Flat year.
2019	39	46.2%	9749	Gold up 10%.
2020	46	58.7%	33609	Gold up 36%.
2021	28	50.0%	357	Gold down 5%.
2022	31	41.9%	447	Gold flat.
2023	33	57.6%	17013	Gold up 12%.
2024	20	80.0%	31972	Outlier; too few trades.

This table explains the final caution. The strategy made money across years, but hit rate was not consistently high. It improved during strong gold years and weakened in flat years. That is exactly what a long-only trend participation strategy should do.

12.7 Risk-Overlay Test Data

The risk overlay was tested with 540 trade anchors and 8000 Monte Carlo paths over a 120-trade horizon in the final optimized artifact. It selected a defensive configuration:

Table 10: Risk-overlay result summary.

Scenario	Benchmark ruin	Optimized ruin	Benchmark MDD95	Optimized MDD95
Baseline	14.35%	0.00%	26.42%	7.65%
Decay	22.85%	0.00%	30.15%	8.54%

The selected configuration used:

- risk fraction per trade 0.0049325843;
- slippage reserve 0.10;
- catastrophe loss cap 0.0105;

- half-size after 3 losses in a 5-trade lookback;
- skip after 4 losses in a 6-trade lookback;
- drawdown ladder at 6%, 10%, 14%, 18%, and 20%.

However, this overlay failed some hard constraints: it did not improve negative-EV cohorts and missed median-PnL constraints. The correct interpretation is that the overlay improves survivability, not predictive edge.

13 Interpretation of the Test Results

13.1 What the Data Supports

The data supports four conclusions:

1. Long-only gold is more robust than symmetric long/short trading in the tested configuration.
2. Trend filtering improves the quality of entries.
3. Realistic forward hit rate is around 48–55%, not 68–75%.
4. Risk overlays can materially reduce ruin and drawdown, but they cannot create missing alpha.

13.2 What the Data Does Not Support

The data does not support:

- a 75% hit-rate claim on current 4H price-only features;
- promotion of in-sample ML top-percentile results;
- promotion of four-trade anomaly results;
- treating 2024's 80% hit rate on 20 trades as a stable property;
- using optimizer survival improvements as proof of predictive edge.

14 What We Learned

1. Price-only 4H XAUUSD features appear to max out near the low-to-mid 50% hit-rate range under honest validation.
2. Long-only trend participation is more robust than symmetric long/short prediction for the tested gold setup.
3. The original 75% target was not impossible in sample, but it did not survive walk-forward validation.
4. The most useful positive signal came from structure that matches gold's macro behavior: trend regime and post-event drift.
5. The 2024 80% hit rate on 20 trades is an outlier, not a stable estimate of forward edge.
6. Risk overlays improve survival and capital preservation, but they cannot turn missing information into predictive alpha.
7. The next genuine edge source is likely event and macro data, not more threshold search on the same price-only feature set.

15 Known Issues and Limitations

Table 11: Known limitations requiring follow-up.

Issue	Why it matters
Kaggle data not independently verified	Research-grade conclusions are acceptable, but production-grade execution requires feed reconciliation and cleaning.
Small validated sample	The pre-registered candidate has 219 trades; this is useful but still vulnerable to regime concentration.
2024 concentration risk	A large portion of headline performance comes from a small number of trades in a favorable gold year.
Price-only feature ceiling	Current features omit the macro variables that directly move gold.
Cost model uncertainty	Reported tests do not fully prove broker-specific spread, slippage, rollover, and liquidity behavior.
4H event timing limitation	Macro shocks can occur and reverse inside a 4H bar; event-aligned testing is required.
Risk overlay overfitting risk	The overlay improved ruin statistics but failed some hard constraints and should be retested on future data.
Conceptual architecture not fully implemented	HMM, continuum-flow variables, and macro forcing are documented as architecture and next-phase direction, not fully validated production machinery.

16 Immediate Next Actions

Table 12: Recommended next actions for Phase 7.

Done	Action	Acceptance criterion
☐	Source macro/event data	FOMC, CPI, NFP, DXY, real-yield, and risk-off series are available with timestamp conventions documented.
☐	Align macro data to 4H bars	Every macro feature is lagged or timestamped so that entries use only information available at decision time.
☐	Test factors individually	Each factor is evaluated in event and non-event cohorts before being combined into a larger model.
☐	Add macro-aware regime model	Regime posterior or admission gate uses macro forcing without lookahead.
☐	Re-run purged walk-forward validation	New model is tested with train/validation/test separation and embargo.
☐	Paper trade current candidate	Current long-only trend-regime strategy is tracked forward with no retrospective parameter edits.
☐	Reconcile data feed	Kaggle-derived bars are compared against the intended broker or institutional feed.

16.1 Decisions Needed

The team should resolve three operational questions before Phase 7 starts:

1. Should the current long-only trend-regime candidate be paper traded while macro-factor research proceeds?
2. What data source should be authoritative for DXY, real yields, CPI, NFP, FOMC timestamps, and XAUUSD execution prices?
3. What minimum forward paper-trading evidence is required before any live-capital discussion?

17 Final Research Conclusion

The mathematical model is useful because it organizes the strategy around state, regime, flow, forcing, risk, and validation. The empirical data is useful because it prevents the model from becoming a story. Together they imply a conservative final position:

The current XAUUSD strategy is a viable paper-trading candidate built around long-only trend-regime gold exposure. The realistic target is 48–55% hit rate with Sharpe 1.5–2.5 if forward performance resembles the pre-registered historical evidence. The original 75% hit-rate target requires event and macro data that the current 4H price-only feature set does not contain.

The next research step is not more parameter searching on the same feature set. It is adding the missing drivers:

$$FOMC_t, \quad CPI_t, \quad NFP_t, \quad \Delta DXY_t, \quad \Delta y_t^{real}, \quad RiskOff_t,$$

then retesting with purged walk-forward validation and explicit event/non-event cohorts.

Source Artifacts

- `reports/phase6_final_methodology_report.md`
- `reports/regime_filtered_BEST/PROPER_BACKTEST_REPORT.md`
- `reports/final_optimized_2026_03_19/optimization_report.md`
- `reports/final_strategy_writeup/FINAL_XAUUSD_STRATEGY_WRITEUP.tex`